

MENTOR CALL

JavaScript Core

Deep dive into modern runtime execution and mechanics

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PART 01

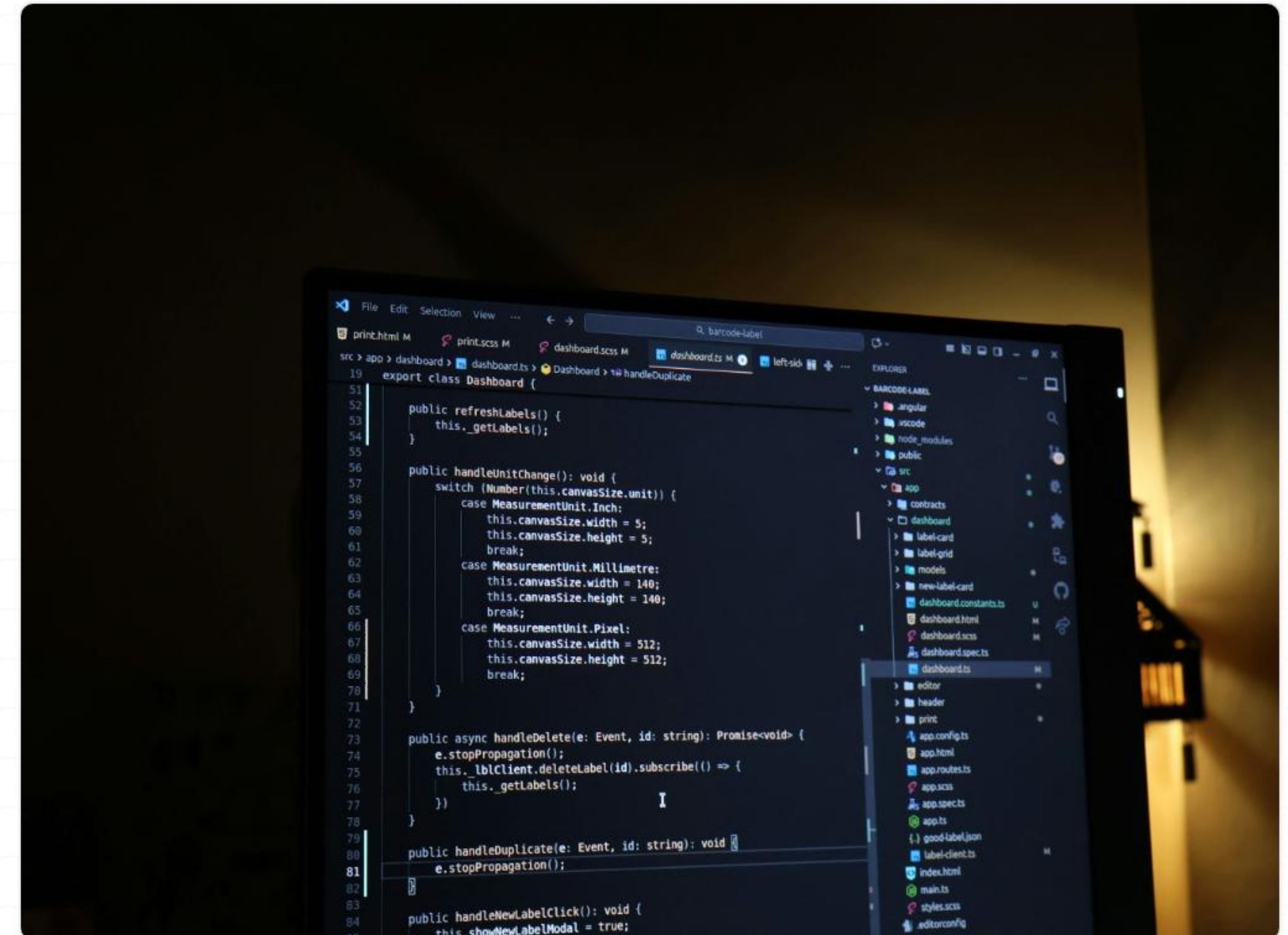
Context & Execution

Exploring variables initialization, memory allocation, and the scoping tree

| Understanding Hoisting

Hoisting is JavaScript's default behavior of moving declarations to the top of the current execution context before execution.

- ❌ Variable Hoisting: Declarations initialized as `undefined`.
- ❌ Temporal Dead Zone: Prevents premature `let` access.
- ❌ Function Hoisting: Complete declaration blocks are accessible.



| Scoping & Execution Blocks



Global Scope

Variables accessible across the entire script environment, declared outside functions.



Function Scope

Variables declared within local bounds. Confines execution environments cleanly.



Block Scope

Modern localized bounds utilizing `let` or `const` inside strict brackets.

PART 02

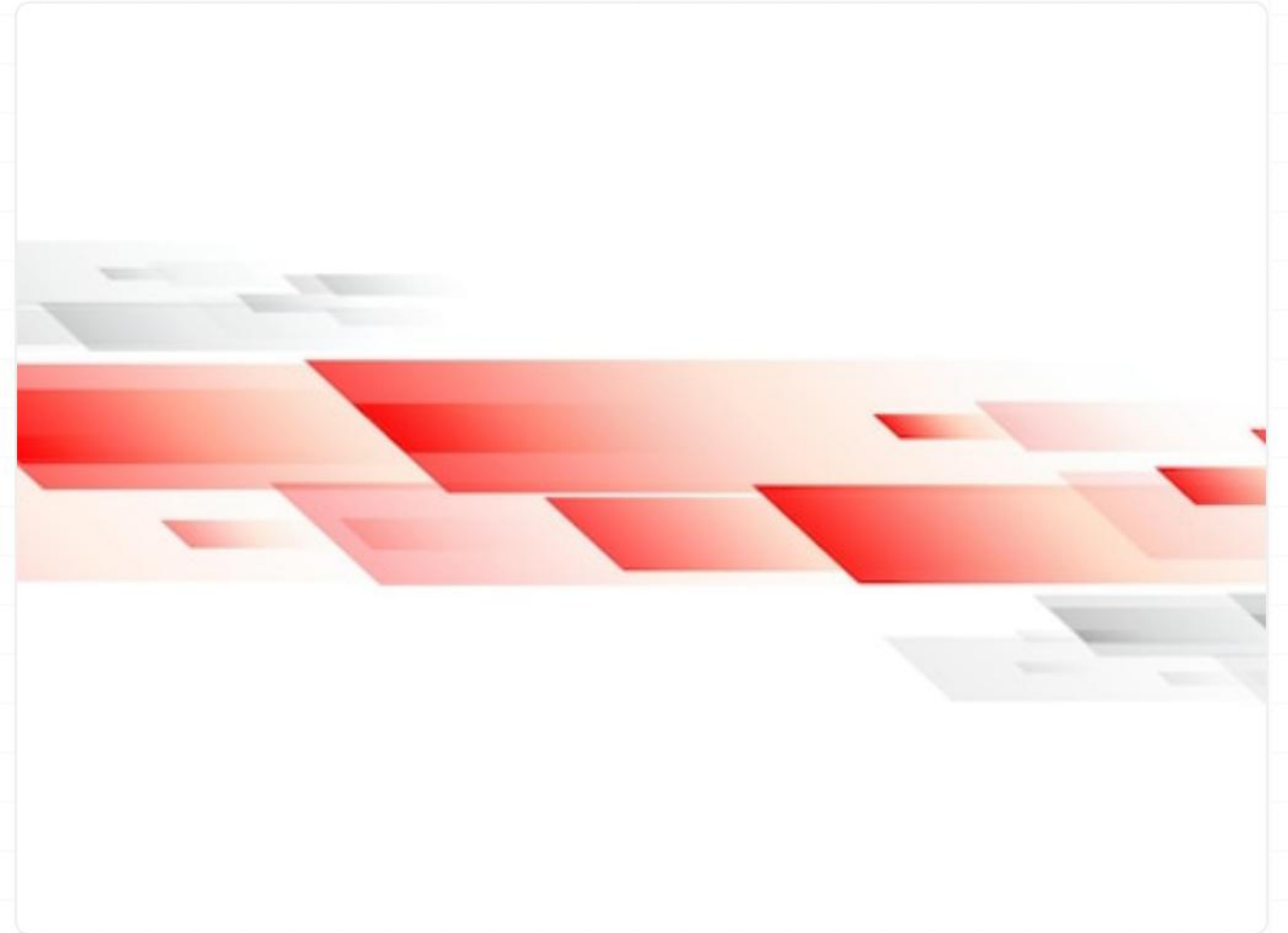
Browser Integration

Interacting with dynamic client environments, arrays, and persistent structures

| Events & Bubbling Flow

Events enable dynamic user input. Bubbling structures propagation sequentially from localized target nodes out to broad parent contexts.

- ❑ Event Handling: Binding functional event listeners.
- ❑ Propagation Flow: Ascends parent hierarchy recursively.
- ❑ Prevention: Halting default behavior with `stopPropagation()` .



| Page Redirection Paths

.href

`window.location.href`

.replace

`window.location.replace()`

.assign

`window.location.assign()`

Saves current state within user history records Clears previous URL entirely from user history Triggers explicit redirection to safe destinations

| Browser Cookies

Cookies store small text files directly inside web client platforms to retain local persistence.

They allow state preservation across session lifecycles, and must be locked down with Secure flags.

| Functional Array Methods

Array Method	Functional Description	Target Output Goal
<code>.map()</code>	Applies a transformation logic cleanly to every index	Returns modified index copies
<code>.filter()</code>	Extracts nested items evaluating against dynamic criteria	Returns exact matching subset
<code>.reduce()</code>	Synthesizes complete array data collections down incrementally	Returns solitary output element

Non-Primitive variables like objects represent dynamic collections of linked key-value descriptors.

- ❑ Reference Value:
References a single memory locus.
- ❑ Mutability:
Structure updates smoothly on the fly.
- ❑ Storage:
Accommodates both parameters and local

Literal Syntax

```
const student = { name: "Alamin", track: "Software Eng", greet: function() { return "Hi!"; } };
```


Questions & Discussion

Interactive debug and concept synthesis call

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| Image Sources



<https://images.unsplash.com/photo-1770734360042-676ef707d022?fm=jpg&q=60&w=3000&auto=format&fit=crop&ixlib=rb-4.1.0&ixid=M3wxMjA3fDB8MHxwaG90by1yZWxhdGVkfDE3fHx8ZW58MHx8fHx8>

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